Jiajun Gao

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EDUCATION

University of Illinois at Urbana-Champaign, Champaign, IL — Fall 2026 - May 2028
- **Master of Science** in Electrical and Computer Engineering

- **Bachelor of Science** in Computer Engineering | **Minor:** Statistics — Aug 2022 - May 2026
- **Undergraduate GPA:** 3.92/4.00 | **Graduated with High Honors**
- **Honors and Awards:** UIUC DGS James Scholar (Spring 2023); UIUC Dean's List (Fall 2022 to Spring 2024, Spring 2025)

PUBLICATION

- Jiajun Gao (2024), "A comparative study between WGAN-GP and WGAN-CP for image generation," *Applied and Computational Engineering*, doi:10.54254/2755-2721/83/2024GLG0059

WORK EXPERIENCE

Shanghai AI Laboratory, Shanghai — Jun 2025 - Sep 2025

Full-Time Algorithm Intern - Intelligent Robot Simulation Platform Based on Isaac Sim and GRUtopia

- Implemented and optimized object and action modules so AlienGo, H1, and Camera Robot platforms could recognize, locate, and orient toward target objects.
- Developed object detection and continuous camera-orientation modules, improving perception stability and interaction precision in complex environments.
- Designed scalable state-transition rules for slicing, microwave, fridge, and transferable-object interactions, removing hard-coded logic.
- Migrated core functionality to Isaac Sim 4.5 and Alibaba Cloud for remote simulation and large-scale testing.
- Integrated Tencent Hunyuan 3D for automatic simulation-object generation, accelerating scene construction and experimentation.
- Designed social-agent preferences and daily-activity workflows to improve realism and scene interactivity.

PROJECT EXPERIENCE

Jarvis Interactive Desktop Companion Robot (ECE 445 Senior Design), UIUC — Spring 2026

Team Leader | Product Concept, AI Integration, Software Development, and Hardware/Software Testing

Recognition: ECE 445 Hall of Fame | Grainger Engineering Senior Design Competition Finalist

- Led a three-person team and originated the concept and end-to-end architecture for an ESP32-S3 desktop companion robot combining conversational AI, expressive feedback, motion, and local safety control.
- Architected and implemented the AI/software pipeline: offline wake-word detection, WebSocket audio streaming, VAD/ASR, Gemini 2.5 Flash response generation, streamed Edge TTS, and MCP-based robot actions.
- Designed the safety-critical control boundary so LLM requests never drive motors directly; implemented ESP32 arbitration, IR obstacle/cliff sensing, motion vetoes, and separate AI and browser-control modes.
- Led hardware/software integration, bring-up, debugging, and validation across custom Face and Body PCBs, LCD expressions, audio, Wi-Fi, motor drivers, battery power, and sensor subsystems.
- Validated a 24-turn session at 2.01 s median and 2.50 s mean ASR-to-first-audio latency with 100% MCP/TTS success; achieved 100% cliff detection across repeated 10-minute motion tests.

Retrieval-Augmented QA under Knowledge Edits | Python, PyTorch, vLLM, Hugging Face, Qwen

- Built a RAG pipeline with BM25 and dense Qwen3-Embedding retrievers and a Qwen2.5-3B generator for multi-hop question answering over a counterfactual knowledge base of **371 edited facts**.
- Implemented cosine-similarity and BM25 retrieval with deterministic top-k ranking invariant to corpus reordering, achieving **87.5% Hit@4**.
- Designed a structured-output prompt that increased answer accuracy from **11.7% to 43.8% (3.7×)**, with **0% JSON parse failures** across **568 generations**.
- Evaluated context sizes (k = 0/1/4/8), finding that excess context reduced accuracy from **43.8% to 36.7%**; diagnosed failures using gold-evidence (oracle) runs and tested robustness to fact edits and answer-order changes.

Autonomous Driving System for GEM Vehicle (ECE 484 Final Project), UIUC Fall 2025

- Built an autonomous driving system on a real GEM vehicle using ROS, integrating perception, control, decision logic, and self-parking.
- Implemented a hybrid Pure Pursuit-Stanley controller with waypoint persistence and linear rate limiting to reduce steering oscillation and cross-track error under GPS noise.
- Developed YOLO-based vision modules for stop-sign, traffic-cone, and gesture detection, enabling safe lane following and vision-triggered self-parking.
- Validated performance on hardware using logged metrics including cross-track error, heading error, steering angle, throttle command, and vehicle speed.
- Recorded GNSS/GPS and IMU orientation topics with rosbag, then post-processed pose traces into a GPS waypoint list for path tracking and parking trajectories.

Loan Approval Prediction (STAT 207 Final Project), UIUC Spring 2025

- Developed a supervised learning model to predict loan approval probabilities using a real-world Kaggle dataset of 4,269 applicants with categorical and numerical variables.
- Completed data cleaning, outlier detection, feature engineering, and exploratory data analysis; identified CIBIL score as the strongest predictor and addressed multicollinearity among numerical variables.
- Selected the most reliable model through backward elimination and 5-fold cross-validation, achieving a mean AUC of 0.962 and a low false-positive rate of 0.0062.
- Proposed regularization and probability calibration schemes to enhance fairness and robustness.
- Delivered a high-accuracy logistic regression model that provided quantitative decision support for loan approvals while revealing key financial risk factors.

RESEARCH EXPERIENCE

Digital Systems Design Lab, UIUC Jan 2025 - May 2025

- Developed FPGA systems in SystemVerilog and Vivado, applying IP-core integration, timing analysis, controller/datapath architecture, and hardware/software co-design.
- Implemented an 8-bit multiplier, SLC-3 microprocessor, AES encryption accelerator, and Nios II-based FPGA soft-CPU SoC platform.
- Co-developed an FPGA-based two-player fighting game with keyboard input, hardware-accelerated sprite rendering, collision detection, and HDMI output using a MicroBlaze CPU and SystemVerilog modules.

A Comparative Study: WGAN-GP vs. WGAN-CP for Image Generation, Shanghai Jul 2024

Team Leader, Summer Research Project | Supervisor: Prof. Mark Vogelsberger, MIT

- Led independent research comparing gradient penalty and weight clipping for Wasserstein GAN image generation.
- Implemented WGAN-GP and WGAN-CP in PyTorch and evaluated stability, convergence, FID, and Inception Score on CIFAR-10 and Fashion-MNIST.
- Authored the published paper, finding that WGAN-GP improved image quality at higher computational cost and clarifying the efficiency-quality trade-off.

COMPETITIONS

- Australian Science Olympiads (ASO) - Physics: Distinction (top 20%)
- Canadian Open Mathematics Challenge (COMC) - Regional Honorable (top 25)
- Math Kangaroo - Proficiency Award (top 25%)

TECHNICAL SKILLS

- **Programming:** C, C++, Python, SystemVerilog, Java
- **Tools & Frameworks:** Git, Linux, ROS, ESP32, MCP, WebSocket, LVGL, Android Studio
- **Data Science & Analysis:** Pandas, NumPy, Matplotlib, Seaborn
- **Machine Learning:** PyTorch, Gemini API, ASR/TTS integration
- **Languages:** Chinese (native), English (proficient)